Computer Aided Engineering Graphics (2025-26)

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Engineering_Graphics

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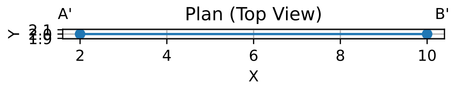
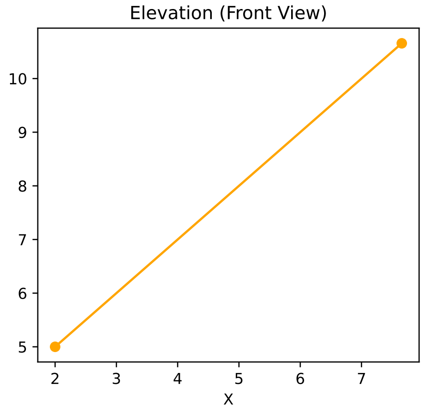
Projection of Lines

```
import matplotlib.pyplot as plt
import numpy as np

# Line parameters
length = 8  # true length of the line
theta = 45  # inclination with VP in degrees

# Plan (top view): line parallel to XY (since parallel to HP)
x_plan = [2, 2 + length]
y_plan = [2, 2]

# Elevation (front view): inclined to XY by theta
x_elev = [2, 2 + length * np.cos(np.radians(theta))]
y_elev = [5, 5 + length * np.sin(np.radians(theta))]

fig, axs = plt.subplots(1, 2, figsize=(10, 4))

# Plan view
axs[0].plot(x_plan, y_plan, marker='o')
axs[0].set_title('Plan (Top View)')
axs[0].set_xlabel('X')
axs[0].set_ylabel('Y')
axs[0].set_aspect('equal')
axs[0].grid(True)
axs[0].annotate('A\'', (x_plan[0], y_plan[0]), textcoords="offset points", xytext=(-10,10), ha='center')
axs[0].annotate('B\'', (x_plan[1], y_plan[1]), textcoords="offset points", xytext=(10,10), ha='center')

# Elevation view
axs[1].plot(x_elev, y_elev, marker='o', color='orange')
axs[1].set_title('Elevation (Front View)')
axs[1].set_xlabel('X')
```

```
Text(0.5, 0, 'X')
```



Plan (Top View)
A'
B'
Y
2.1
2.0
1.9
2
4
6
8
10
X



Elevation (Front View)
10
9
8
7
6
5
2
3
4
5
6
7
X

Chapter 3

Projection of Planes

3.1 Introduction

- A plane is a two-dimensional surface that has only length and breadth; its thickness is considered negligible.
- In engineering graphics, planes are depicted using various geometric shapes, including:
  - Squares
  - Rectangles
  - Circles
  - Pentagons
  - Hexagons
  - Other polygons
- These plane figures form the foundation for understanding and visualizing the projection of objects onto different reference planes.

3.2 Typical Problem Statements

- What is usually asked in problems?
  - To draw the projections:
    - Front View (F.V.)
    - Top View (T.V.)
    - Side View (S.V.)
- What information will be given in the problem?
  - The type/description of the plane figure (triangle, square, hexagon, etc.).
  - Its position with respect to Horizontal Plane (H.P.) and Vertical Plane (V.P.).
- How will the position with H.P. & V.P. be described?
  - The incination of the surface/ lamina of the plane with one of the reference planes is given.

– The position/inclination of one of its corners/edges with repect to the other reference plane.
– This means the plane, we consider, is usually inclined to both reference planes.

3.3 Stages for Drawing Projections When a Plane is Inclined

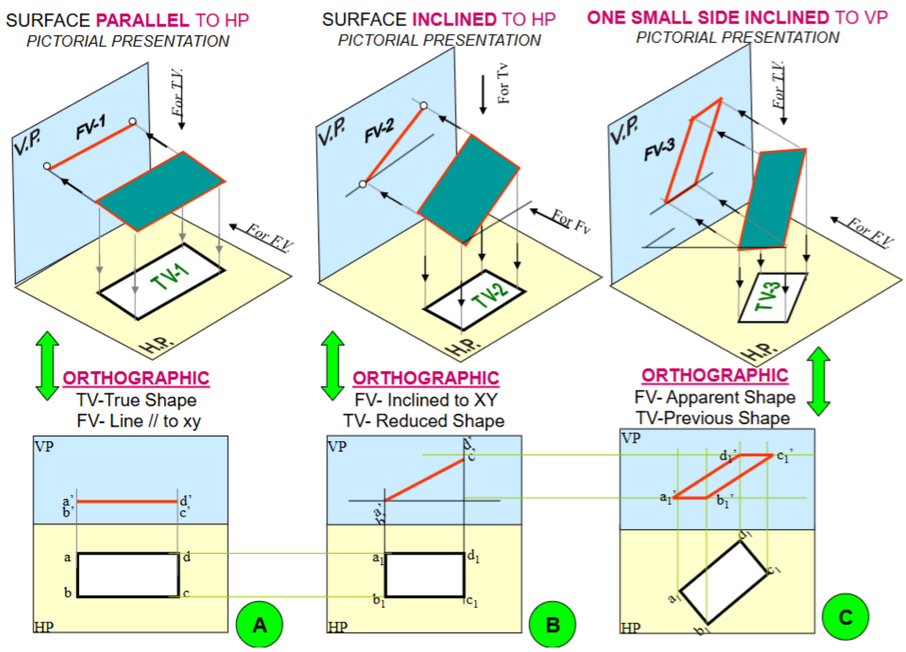


Figure 3.1: Stages of Drawing Projections for an Inclined Plane.
Image reference: https://iitg.ac.in/kpmech

The process starts by assuming an initial simple position, then introducing the surface inclination, and finally introducing the side/edge inclination.

- Initial Position: Surface Parallel to one Reference Plane
  - Assumption: The plane figure is initially assumed to be parallel to one of the reference planes (either HP or VP). This allows one of its views to show its true shape.
  - Views:
    * If the surface is assumed parallel to HP, the Top View (TV) will show the true shape of the plane figure. The Front View (FV) will appear as a line parallel to the XY line.
    * If the surface is assumed parallel to VP, the Front View (FV) will show the true shape of the plane figure. The Top View (TV) will appear as a line parallel to the XY line.

- Surface Inclination
  - Description: The inclination of the plane's surface (lamina) with one of the reference planes is introduced in this step.
  - Transformation: The view that was a line in the previous step (e.g., FV if the surface was parallel to HP) will now be drawn inclined to the XY line at the given angle. The other view (e.g., TV) will show a reduced shape.
- Edge/Side Inclination with the Other Reference Plane
  - Description: The inclination of one of the plane's edges (or a specific feature like a diameter or diagonal) with the other reference plane is introduced. This leads to a case where the object is inclined to both reference planes.
  - Transformation: The view from the previous step that showed the "reduced shape" (e.g., TV in Problem 1) is now redrawn such that the specified edge or feature makes the given angle with the XY line (representing the other reference plane). The final views will show apparent shapes.

How to Approach a Projection Problem

Read the problem and answer the following questions:

1. Surface inclined to which plane? (`$ \color{blue}{\mathrm{H.P.}} $` or `$ \color{blue}{\mathrm{V.P.}
2. Assumption for initial position? — **Parallel to ?**
3. So which view will show the true shape? — **____ View**
4. How to draw the edge/ corner which touches one of the planes? — **____**